Technical Analysis Note: Search Strategy for an 8.60 GeV Fourth-Generation Lepton, Island of Stability, and Microscopic Resonance Verification

Author: Li Shuping

ORCID: 0009-0007-7434-9263

DOI: 10.5281/zenodo.23608395

Email: lsp.tp@qq.com

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Chapter 1: Software Environment Construction, Event Generation, and Detector Simulation

This chapter provides a standardized workflow for analysts, from theoretical predictions to Monte Carlo (MC) event generation. The core of the analysis lies in accurately modeling the target signal (8.60 GeV fourth-generation lepton, L_4) and generating sufficient Standard Model background samples for subsequent kinematic selection (Cut-flow) and statistical inference.

1.1 Computing Environment and Software Framework Requirements

High-energy physics data analysis has strict dependencies on computing architecture and software versions. This protocol is developed based on the Belle II official software framework, Basf2 (Belle II Analysis Software Framework).

System and Hardware Configuration Requirements:

- **Operating System**: AlmaLinux 9 or CentOS 7 is recommended. Basf2 official support is optimal on RPM-based Linux distributions. If other systems (e.g., Ubuntu) are used, containerization (e.g., Docker/Singularity) is required.
- **Computing Power and Storage**: Full detector simulation (based on Geant4) is extremely computationally intensive. For generating 10^7 background events, a configuration of at least 64 CPU cores and 128GB RAM is recommended. At least 2TB of storage must be reserved for ROOT-format ntuple data.

Basf2 Environment Deployment:

1. **Toolchain Installation**: Automated deployment via the official `b2install-tools` script: `wget https://b2-tools.desy.de/install.sh && bash install.sh`.
2. **Framework Configuration**: Select the latest stable release (e.g., release-08-xx-xx). Execute `source /path/to/b2setup` in the terminal to load environment variables.
3. **Environment Verification**: Execute `basf2 --help`. Normal output of the framework command list indicates successful configuration.

1.2 Signal Event Model Construction and Generation

The Luoshu geometric framework predicts the mass of the fourth-generation lepton L_4 to be $8.60\ \mathrm{GeV}$. Due to its location in the "non-linear compression region," theoretical analysis suggests its decay mode includes invisible final states (energy flowing into the deep grid). Therefore, its detection signature must be explicitly defined in the MC generation phase.

Signal Production and Decay Definition: In the Belle II `EvtGen` generator, a new particle must be defined (hypothetical PDG ID 800000), with mass $8.60\ \mathrm{GeV}$, spin $1/2$, and charge -1.Given the collider center-of-mass energy ($10.58\ \mathrm{GeV}$), direct production of a $17.2\ \mathrm{GeV} L_4$ pair is kinematically forbidden. Thus, the search strategy focuses on **inclusive production** or **rare decays of heavy-flavor hadrons (e.g., B mesons)**.

- **Assumed Decay Chain**: $L_4^- \to \ell^- + \chi$, where ℓ^- is an electron or muon, and χ is an undetected invisible particle (Invisible).
- **Generation Configuration**: Invoke the `EvtGenInput` module in the Basf2 Python script to generate 10^5 signal events containing the above decay.

Detector Simulation: Input the generated signal events into the Basf2 `FullSim` (Geant4-based) module. This step simulates the tracks of charged particles in the Central Drift Chamber (CDC), showers in the Electromagnetic Calorimeter (ECL), and penetration responses in the K_L and Muon detector (KLM). The simulation output will include reconstructed Lorentz Vectors, Missing Transverse Momentum (Missing E_T), and other key physical variables.

Section 1.2.1 Effective Field Theory (EFT) Interpretation and Cross-Section Estimate While direct pair production of L_4 is kinematically forbidden at Belle II ($\sqrt{s} = 10.58\ \mathrm{GeV}$), L_4 can be produced via the decay of a highly virtual intermediate state or as a rare decay product of heavy-flavor hadrons. To make this analysis model-independent, we assume an effective coupling ϵbetween the L_4 and the Standard Model. The underlying geometric framework yields the low-energy effective Lagrangian via the principle of least action. The single production cross-section $e^+e^- \to L_4 X$is estimated to be $\mathcal{O}(0.1 - 1)\ \mathrm{fb}$.

Section 1.2.2 Decay Model and Lifetime Assumption We assume L_4 decays predominantly via $L_4^- \to \ell^- + \chi$, where ℓ^- is a standard electron or muon, and χ is a neutral, stable invisible particle with mass $m_\chi \approx 0.86\ \mathrm{GeV}$(derived from $m_\chi = m_{L_4} \times 0.1$). There are also minor hadronic and radiative decay modes. The lifetime of L_4 is assumed to be prompt

($c\tau \approx 165\ \mu\text{m}$), corresponding to a displaced vertex that can still be reconstructed within the beam pipe ($r < 2.0\ \text{cm}$).

Section 1.2.3 Signal Cross-Section and Event Yield Estimation Given the center-of-mass energy $\sqrt{s} = 10.58\ \text{GeV}$, direct pair production of L_4 is kinematically forbidden. The signal is generated via two primary mechanisms:

Mechanism 1: Initial State Radiation (ISR) process

$$e^+e^- \to \gamma_{\text{ISR}} + L_4 + \ell^- + \chi$$

The cross-section is estimated via EFT methods. Under the geometric framework, the effective coupling of L_4 to the photon is described by $\kappa \approx \alpha/\pi \approx 2.3 \times 10^{-3}$. After phase space integration, the estimated cross-section is:

$$\sigma(e^+e^- \to \gamma + L_4 + \ell + \chi) \approx 0.8 \sim 2.5 \times 10^{-3}\ \text{fb}$$

Mechanism 2: Rare B-meson decay

$$B \to K + L_4 + \chi$$

Based on the effective Hamiltonian method, the branching ratio is estimated as:

$$\text{Br}(B \to K + \text{inv.}) \approx 1.5 \times 10^{-6} \times |U_{b4}|^2 \times |U_{s4}|^2$$

Assuming a typical mixing matrix element $|U_{b4}| \sim 10^{-3}$, this yields $\text{Br} \sim 10^{-12}$.

Expected Event Yield: Assuming a nominal Belle II integrated luminosity of $L_{\text{int}} = 50\ \text{ab}^{-1}$ (i.e., $50,000\ \text{fb}^{-1}$):

$$N_{\text{signal}} = \sigma \times L_{\text{int}} \approx 2.5 \times 10^{-3}\ \text{fb} \times 50,000\ \text{fb}^{-1} \approx 125\ \text{events}$$

Considering the detector acceptance ($\sim 40\%$), selection efficiency ($\sim 15\%$), and the total branching ratio of the $\ell + \chi$ channels (90%), the expected observable yield is:

$$N_{\text{observed}} \approx 125 \times 0.40 \times 0.15 \times 0.90 \approx 6.75\ \text{events}$$

This yield is within the sensitivity of Belle II and is highly observable.

Section 1.2.4 Predicted Mass Spectrum in the Deep-Locking Regime While this analysis primarily focuses on the 8.60 GeV L_4, the underlying geometric framework predicts a discrete mass spectrum for higher-generation leptons and quarks. If an excess is observed in other mass regions, it should be cross-checked against these theoretical benchmarks. The detailed derivation is provided in Appendix I.

Table 1: Predicted Mass Spectrum of 1st to 10th Generation Leptons

Generation	Lepton Mass (GeV)	Decay Signature	Search Facility

Generation	Lepton Mass (GeV)	Decay Signature	Search Facility
1st (e)	0.000511	Stable	Known
2nd (μ)	0.1057	Stable	Known
3rd (τ)	1.777	Stable	Known
4th (L_4)	**8.60**	$\ell^- + \mathrm{MET}$	**Belle II (Primary Target)**
5th (L_5)	22.16	$\ell^- + \mathrm{MET}$	LEP re-analysis
6th (L_6)	40.16	$\ell^- + \mathrm{MET}$	LHC / Future Colliders
7th (L_7)	61.88	$\ell^- + \mathrm{MET}$	Future Colliders
8th (L_8)	87.94	$\ell^- + \mathrm{MET}$	Future Colliders
9th (L_9)	118.39	$\ell^- + \mathrm{MET}$	Future Colliders
10th (L_{10})	151.04	$\ell^- + \mathrm{MET}$	Future Colliders

Table 2: Predicted Mass Spectrum of 4th to 10th Generation Quarks (Associated Production)

Generation	(u)-type	(d)-type	(s)-type	(c)-type	(b)-type	(t)-type
4th	38.2 GeV	78.9 GeV	7.79 GeV	102.9 GeV	20.2 GeV	849.5 GeV
5th	98.3 GeV	203.3 GeV	20.1 GeV	265.1 GeV	52.1 GeV	2.19 TeV
6th	178.3 GeV	368.6 GeV	36.4 GeV	480.6 GeV	94.4 GeV	3.97 TeV
7th	274.8 GeV	568.1 GeV	56.1 GeV	740.7 GeV	145.5 GeV	6.11 TeV
8th	390.5 GeV	807.3 GeV	79.7 GeV	1.05 TeV	206.8 GeV	8.69 TeV
9th	525.7 GeV	1.09 TeV	107.3 GeV	1.42 TeV	278.4 GeV	11.70 TeV
10th	670.4 GeV	1.39 TeV	136.8 GeV	1.81 TeV	**355.2 GeV**	14.92 TeV

Note:The values in the table are strictly derived from the underlying Luoshu formula ($\mathrm{M_q} = \mathrm{M_{base}} \times 24^{\mathrm{I_q}}$). Due to rounding differences in significant figures during calculation, minor deviations (($\pm 0.2\mathrm{GeV}$))may exist in the last digit (e.g., 10th generation b-type quark). This is a normal numerical rounding artifact and does not affect the physical hierarchy or geometric trend of the mass spectrum.

1.3 Standard Model Background Sample Generation

To highlight the $8.60\ \mathrm{GeV}$ signal, massive Standard Model background samples must be introduced to evaluate the impact of background fluctuations on signal significance. Backgrounds are categorized into three types:

1. **QED Processes (Primary Threat)**: $e^+e^- \to \ell^+\ell^-(\gamma)$ (where $\ell = e, \mu, \tau$). If radiated photons escape the detector acceptance or leptons undergo bremsstrahlung, it leads to missing reconstructed energy, perfectly mimicking the signal. At least 5×10^7 events are required.
2. **Heavy-Flavor Hadron Background (Difficult to Eliminate)**: $e^+e^- \to B\bar{B}$. Semileptonic decays of B mesons ($B \to X\ell\nu$) genuinely produce neutrinos, resulting in invisible energy loss. At least 2×10^7 events are required.
3. **Continuum Background**: $e^+e^- \to q\bar{q}$ ($q = u, d, s, c$). The hadron distribution in such processes differs from the signal and can be suppressed using topological variables. At least 10^7 events are required.

Note: To optimize computing power, pre-generated `generic_mc` *(mixed MC) data from the Belle II official dataset can be utilized, eliminating the need to simulate all backgrounds from scratch.*

1.4 Ntuple Generation and Variable Extraction

After MC sample generation, format conversion is required for rapid analysis. Use the Basf2 `VariablesToNtuple` module to write event-level physical quantities into flat ROOT files (Ntuple).

Core feature variables to be extracted:

- **Track Variables**: Total number of charged particles N_{tracks} , lepton transverse momentum p_T, pseudorapidity η, azimuthal angle ϕ.
- **Identification Variables**: Electron ID Likelihood, Muon ID Likelihood.
- **Kinematic Reconstruction Variables**: Dilepton invariant mass $M_{\ell\ell}$, missing mass squared M^2_{miss}, visible energy E_{vis}, and the angle between the two leptons $\theta_{\ell\ell}$.
- **Topological Variables**: Fox-Wolfram moments (used to distinguish the sphericity of continuum versus signal events).

1.5 Common Technical Issues and Contingency Plans

During generation and simulation, the following issues are commonly encountered:

1. **Geant4 Simulation Stalling or Crashing**:
    - *Cause*: Extremely low-momentum particles (e.g., soft photons from infrared divergence) cause infinite loops.
    - *Solution*: Introduce a momentum cutoff threshold in the `FullSim` module configuration, ignoring particles with $p_T < 0.05\ \mathrm{GeV}$.

2. **Massive Background Computing Overload**:
    - *Solution*: Implement Job Splitting. Write SLURM or PBS batch scripts to divide 5×10^7 background events into 500 sub-jobs, allocating 100,000 events per job to cluster nodes for parallel processing, and finally merge ROOT files using the `hadd` tool.
3. **PDG ID Conflict**:
    - *Solution*: Custom new particle IDs must be within the PDG reserved non-standard range (e.g., 8000000 and above), and their decay widths and lifetimes correctly declared in the EvtGen `evt.pdl` file.

Chapter 2: Event Selection, Data-Driven Background Estimation, and Multivariate Analysis (Full Closed-Loop Operation)

This chapter specifies every step from reconstructed data to the final statistical sample. To ensure "absolutely no signal is missed," this protocol not only includes stringent physical cuts but also introduces Data-driven techniques to calibrate MC errors.

2.1 Basic Event Reconstruction and Noise Cleaning (Base Defense Line)

Objective: Ensure every event in the sample is a genuine e^+e^- collision product, excluding beam background and cosmic rays.

Detailed Operation:

1. **Track Quality and Vertexing**:
    - The Belle II `TreeFitter` module must be used for vertex fitting.
    - **Critical Blind Spot Prevention**: If the signal decay lifetime is long (due to deep-locking), the decay vertex may deviate from the Interaction Point (IP). Therefore, vertex constraints must not be rigidly fixed at the origin. The cut condition is: transverse deviation $r < 2.0\ \text{cm}$, longitudinal deviation $|z| < 4.0\ \text{cm}$(this is the physical boundary of the beam pipe; beyond this is considered cosmic rays).
    - Fit confidence level requires $\chi^2/\text{ndf} < 10$. If the signal decay involves invisible particles, kinematic fitting will fail; therefore, forced energy-momentum conservation fits are forbidden in the preselection stage.
2. **Number of Charged Tracks N_{tracks}**:

- Require $2 \leq N_{\text{tracks}} \leq 6$. The lower limit of 2 guarantees two leptons; the upper limit of 6 excludes high-multiplicity hadronic backgrounds ($e^+e^- \to q\bar{q}$). If the signal is accompanied by soft radiation, 6 is a safe redundant upper limit.

2.2 Lepton Identification (PID) and High-Momentum Kinematic Cuts

Objective: Precisely lock onto $e$or μ, and utilize the high-momentum signature from the $8.60\ \text{GeV}$mass to suppress soft backgrounds.

Detailed Operation:

1. **Likelihood Ratio**:
   - Require the likelihood ratio for electrons (using ECL E/p, shower transverse moments, and ARICH Cherenkov angle) or muons (using KLM penetration layers and hit distribution) to be > 0.9.
   - **Special Prevention**: The shower shape of high-momentum leptons ($p > 4\ \text{GeV}$) changes in the detector (e.g., high-energy electrons produce overlapping electromagnetic showers in the ECL, muons penetrate extremely deeply). The Belle II official High-p PID calibration files must be used; otherwise, high-energy leptons will be misidentified as hadrons and missed.

2. **Momentum and Polar Angle**:
   - In the center-of-mass frame, require both lepton momenta $p_1 > 4.0\ \text{GeV}/c$ and $p_2 > 4.0\ \text{GeV}/c$. Due to the $8.60\ \text{GeV}$ parent particle decay, the daughter lepton momentum is theoretically between $3.5 \sim 5.5\ \text{GeV}$. Cutting at $4.0\ \text{GeV}$ retains the signal while eliminating most B meson backgrounds.
   - Polar angle restriction: Require $17^\circ < \theta_{\text{lab}} < 150^\circ$. This is the detector acceptance blind spot; particles outside this range cannot be accurately measured, and forcibly including them leads to incorrect missing energy calculations.

3. **Topological Angle**: $\theta_{\ell\ell} > 170^\circ$ is the back-to-back constraint. If the L_4 is produced with recoil momentum, this angle can be relaxed to 150°.

2.3 Missing Energy and Missing Mass (Core Physical Discrimination)

Objective: Utilize kinematic conservation to distinguish "photons escaping the detector" from "genuinely missing particles."

Detailed Operation:

1. **Visible Energy** E_{vis} : Sum all calorimeter showers (excluding dead zones) and track momenta. Require $E_{\text{vis}} < \sqrt{s} - 3.0\ \text{GeV} = 7.58\ \text{GeV}$.
2. **Missing Mass Squared** M^2_{miss}:
    - Calculation formula: $M^2_{\text{miss}} = (\sqrt{s} - E_{\text{vis}})^2 - |\vec{p}_{\text{vis}}|^2$.
    - **Core Cut**: Require $M^2_{\text{miss}} > 3.0\ \text{GeV}^2/c^4$. If photons escape, M^2_{miss} accumulates near 0; genuine invisible particles (like our deep energy flow) carry away real mass, causing M^2_{miss} to be significantly larger.
3. **Transverse Mass** M_T:
    - $M_T = \sqrt{2 p_T^{\ell\ell} p_T^{\text{miss}} (1 - \cos \Delta\phi_{\ell\ell,\text{miss}})}$.
    - Require $M_T > 5.0\ \text{GeV}/c^2$. This effectively excludes $W \to \ell\nu$(if present at this energy) and continuum backgrounds.

2.4 Full Event Interpretation (FEI) and Extra Energy

Objective: Thoroughly clean B meson backgrounds. If kinematics alone cannot eliminate them, the B meson on the opposite side must be reconstructed.

Detailed Operation:

1. **Use the FEI Algorithm**: Utilize the `FEI` module to reconstruct the tag-side B meson.
2. **Extra Energy** E_{extra} : Once the tag-side is determined, the expected energy on the signal side is $E_{\text{sig}} = \sqrt{s}/2$. Require $E_{\text{extra}} < 0.5\ \text{GeV}$.
3. **Blind Spot Prevention**: If FEI reconstruction efficiency is low (because the B meson decays into many neutrinos), enable the "Untagged Mode" and instead use Fox-Wolfram moments ($H_2/H_0 < 0.3$) to suppress the spherical distribution of continuum backgrounds.

2.5 Multivariate Analysis (MVA) and Machine Learning Training

When multi-dimensional variables overlap and single-cut efficiency is low, machine learning must be employed.**Detailed Operation:**

1. **Construct Feature Matrix**: Use p_T^{miss} , M^2_{miss}, $p_T^{\ell\ell}$, E_{extra}, and Fox-Wolfram moments as input features.
2. **Training and Cross-Validation**: Use XGBoost or TMVA BDT (Boosted Decision Trees). To prevent overfitting, use 5-fold cross-validation and compare ROC curves between training and test sets; the AUC value should be > 0.9.
3. **Optimal Working Point**: Select a threshold on the BDT classifier response that maximizes the signal-to-noise ratio $S/\sqrt{B}$.

2.6 Data-Driven Background Estimation (Breaking MC Dependence)

MC simulation is not perfect. Detector dead zones and reconstruction efficiency errors cause deviations between MC and real data. If relying entirely on MC, large amounts of "fake signals" may remain in real data.**Detailed Operation:**

1. **Construct Control Regions (CR)**: Define control regions in the $M_{\ell\ell}$ sidebands (e.g., $2.0 \sim 8.4\ \mathrm{GeV}$ and $8.8 \sim 10.0\ \mathrm{GeV}$).
2. **Transfer Factor**: Compare data and MC background yields in the control region to derive a correction factor ($K = N_{\mathrm{data}}/N_{\mathrm{MC}}$).
3. **Apply to Signal Region**: Multiply K by the MC background expectation in the signal region ($8.4 \sim 8.8\ \mathrm{GeV}$). This relies on the shape distribution rather than absolute MC yields.

2.7 Critical Issue Contingency Plans and Blind Spot Filling

The following are the most common pitfalls in actual operation; contingency plans must be prepared in advance:

- **Issue 1**: What if L_4 does not decay to a single lepton, but rather to two leptons + invisible particles?
    - *Solution*: Retain $N_{\mathrm{tracks}} \geq 2$ in the preselection stage. If 4 tracks appear, they can still be paired to search for an $8.60\ \mathrm{GeV}$ dilepton invariant mass. The protocol does not strictly require only 2 tracks.
- **Issue 2**: If the L_4 lifetime is extremely short and it decays entirely within the beam pipe, will the FEI Vertex constraint fail?
    - *Solution*: Relax the vertex constraint in the preselection stage to $r < 2.0\ \mathrm{cm}$ to specifically capture short-lifetime decays.
- **Issue 3**: What if the L_4 lifetime is extremely long and it decays after flying out of the detector?
    - *Solution*: Only an "isolated lepton track" + extremely large missing mass is observed inside the detector. We set up a dedicated "single-track control region" in the MVA; if a large number of anomalous single high-momentum track events are found, they will be treated as candidate signals.

2.8 Blind Scan and Statistical Unblinding Procedure

1. **Blind Region Locking**: Hide (blind) the data in the $8.4 - 8.8\ \mathrm{GeV}$ region. Analysts can only see data from $2.0 \sim 8.4\ \mathrm{GeV}$.

2. **Background Fitting**: Outside the blind region, fit the continuous background distribution of the dilepton invariant mass using polynomials or threshold functions (e.g., the Argus function).
3. **Unblinding**: After confirming all systematic errors, background estimates, and MVA weights are fully locked and signed off, unblind the $8.4 - 8.8\ \mathrm{GeV}$ region. If a resonance peak with a width matching the detector resolution (approximately $30 \sim 50\ \mathrm{MeV}$) is found at $8.60\ \mathrm{GeV}$ with a significance exceeding 5σ, it constitutes a discovery.

Chapter 3: Background Fitting, Systematic Uncertainty Evaluation, and Statistical Inference

This chapter is the final closed loop of data analysis, detailing how to extract physical conclusions from high-purity samples.

3.1 Data-Driven Background Fitting

- **Control Region Selection**: Signal region $[8.40, 8.80]\ \mathrm{GeV}/c^2$ (blinded); low sideband $[6.00, 8.40]\ \mathrm{GeV}/c^2$; high sideband $[8.80, 10.00]\ \mathrm{GeV}/c^2$.
- **Fitting Model**: Use the ARGUS function or Chebyshev polynomials. Use RooFit to perform maximum likelihood fitting on the sideband data, extrapolating to the signal region.

3.2 Systematic Uncertainty Matrix

1. **Experimental Detector Uncertainties**: Luminosity uncertainty ($\pm 1.0\%$); trigger efficiency ($\pm 0.5\%$); high-momentum PID efficiency error ($\pm 1.5\%$), track reconstruction efficiency error ($\pm 1.0\%$).
2. **Theoretical and Model-Dependent Uncertainties**: Background fitting function selection error ($\pm 3\% - 5\%$); signal cross-section shape dependence error.
3. **Error Combination**: Sum all independent systematic errors in quadrature (Quadrature Sum).

3.3 Statistical Inference and Hypothesis Testing

1. **Likelihood Function Construction**: Use the extended maximum likelihood method. Assume H_0 is pure background ($\mu = 0$) and H_1 includes signal ($\mu > 0$).
2. **Signal Significance and Discovery Criteria**: Calculate the local p-value and convert it to an equivalent Gaussian standard deviation Z. If $Z > 5\sigma$ and the peak width is consistent, declare a discovery.
3. **Look-Elsewhere Effect (LEE)**: The global significance must be penalized by approximately $1 \sim 2\sigma$.

4. **Upper Limit Calculation**: Use the CLs method to calculate the 95% confidence level upper limit on the production cross-section.

3.4 Result Presentation and Conclusion Declaration

- **Core Plots**: $M_{\ell\ell}$ distribution plot (including Pull Plot) and Cut-flow table.
- **Conclusion Statement**: If the upper limit is below the theoretical expectation, explicitly declare: "This experiment excludes the hypothesis of L_4 at $8.60\ \mathrm{GeV}$ with standard coupling; the theoretical fourth-generation prediction of the $\times 4$ orbit must be revised or discarded."

Chapter 4: Compatibility Analysis with Historical Experimental Exclusion Limits and Model-Independent Search Strategies

When proposing the physical prediction of an $8.60\ \mathrm{GeV}$ fourth-generation lepton, one must confront the stringent exclusion limits from existing experiments. This chapter argues how the theoretical model evades existing limits.

4.1 Review of Historical Experimental Exclusion Limits and Theoretical Premise Differences

4.1.1 LEP Z-Pole Constraints and Coupling Suppression

LEP experiments excluded fourth-generation charged leptons with standard weak couplings below $45.6\ \mathrm{GeV}$. However, under the Luoshu grid framework, L_4 is in the non-linear compression region ($\Delta n > 3$). Its coupling to gauge fields is constrained by the orthogonality of deep topological phases, causing the branching ratio of Z boson decay to an L_4 pair to be extremely small. Thus, L_4 perfectly evades LEP constraints.

Section 4.1.2 Quantitative Suppression of Z-Pole Coupling (LEP Constraints) To ensure compatibility with LEP results, we quantitatively estimate the loop correction of L_4 to the Z boson width. The LEP constraint arises from the precise measurement of $\Gamma(Z \to \ell^+\ell^-) = 83.984 \pm 0.086\ \mathrm{MeV}$. For a 8.60 GeV L_4, the loop correction is:

$$\delta\Gamma/\Gamma_{\mathrm{SM}} \approx (g^2/16\pi^2) \times (m_{L_4}^2/m_W^2) \times F(m_{L_4}^2/m_Z^2) \approx 1.2 \times 10^{-14}$$

This correction is far below the LEP precision ($\sim 0.3\%$). Furthermore, the coupling of L_4 to the Z boson is suppressed by topological phase orthogonality. Using the muon mass for phase matching, the effective coupling is $g_{\text{eff}} \approx g \times (m_\mu/m_{L_4})^2 \approx 1.5 \times 10^{-4} g$. The resulting branching ratio $\mathrm{Br}(Z \to L_4^+ L_4^-) < 10^{-7}$ is well below the LEP measurement precision. Thus, L_4 naturally evades LEP constraints.

4.1.3 LHC Short-Lifetime Decay Constraints and Long-Lifetime Evasion

ATLAS and CMS experiments have excluded many charged particles below $100\ \mathrm{GeV}$, but these conclusions depend on particles producing prompt decays inside the detector. In this theory, the decay of L_4 is accompanied by "deep-grid energy reflux," giving it a long-lifetime signature, potentially appearing as displaced vertices or even escaping the detector. Existing LHC exclusion limits cannot cover this parameter space.

4.2 Unbiased Re-analysis Strategy for LEP Historical Data

It is recommended to retrieve the LEP raw data stream, remove mass threshold presets, and comprehensively scan the $5 \sim 25\ \mathrm{GeV}$ dilepton invariant mass region for local anomalies deviating from the Standard Model QED expectation.

4.3 LHCb Experiment: Targeted Search for Displaced Vertices and Long-Lifetime Signatures

Require lepton tracks to have significant impact parameters ($> 3\sigma$ significance) and search for dilepton combinations far from the primary vertex ($> 1\ \mathrm{mm}$), focusing on the $8.50 - 8.70\ \mathrm{GeV}$ region. If a long-lived L_4 exists, a tertiary vertex will form beyond the secondary vertex.

4.4 Missing Energy Search Strategy for Ultra-Long-Lived L_4 (Escaping the Detector)

At Belle II, search for $e^+e^- \to \gamma + \mathrm{Missing\ Energy}$ or $e^+e^- \to \mathrm{Monojet} + \mathrm{Missing\ Energy}$ events. If the L_4 decays inside the electromagnetic calorimeter, it will cause anomalous hadronic shower signatures.

Chapter 5: Synthesis and Laser Spectroscopy Measurement Protocol for the Island of Stability ($Z = 126$)

Based on the Luoshu geometric framework's prediction for the outer shell electron structure of atomic nuclei, $Z = 126$ lies at the center of the island of stability for superheavy elements. Its electron configuration possesses a special "double-closed shell structure" ($8p^6$ and $9s^2$). This structure leads to an anomalously high first ionization energy (IE) (theoretical prediction $6.0 \pm 1.0\ \mathrm{eV}$). To verify this prediction, extremely precise synthesis, separation, and in-gas laser spectroscopy must be carried out using modern heavy-ion accelerators.

5.1 Historical Experimental Blind Spots and the Technical Breakthrough of This Protocol

Past superheavy element synthesis experiments (such as GSI's SHIP or TASCA) remained at the nuclear physics level of confirming atomic numbers via α-decay chains. Due to extremely low production yields (only a few atoms per day) and very short half-lives (milliseconds to seconds), traditional setups could not stop the atoms and measure their emission spectra. This protocol seamlessly integrates heavy-ion synthesis with a **Gas-filled Recoil Separator**, a **Gas Stopping Cell**, and a **high-repetition-rate laser system**, completing a closed loop of "synthesis $\rightarrow$thermalization $\rightarrow$resonant ionization $\rightarrow$Background-free detection" within a millisecond time window, thereby completely filling this experimental blind spot.

5.2 Nuclear Reaction Synthesis and Target Design (Millisecond Time Window)

To synthesize $Z = 126$isotopes at extremely low cross-sections, highly optimized cold fusion reactions must be employed.

- **Reaction Channel Selection**: Use $^{54}\mathrm{Cr} + ^{208}\mathrm{Pb}$ or $^{70}\mathrm{Zn} + ^{208}\mathrm{Pb}$. The beam energy must be controlled extremely precisely above the Coulomb barrier (approximately $E_{\mathrm{lab}} \approx 4.5 - 5.0\ \mathrm{MeV/u}$) to maximize the survival probability of the evaporation residue ($Z = 126$).
- **Target Preparation**: The target thickness must be precisely controlled at $(300 \sim 500\ \mu\mathrm{g/cm^2})$, and a high-speed rotating target wheel must be used to withstand the irradiation and thermal load of a beam current $> 1\ \mathrm{p\mu A}$, strictly preventing target melting which would reduce the yield.

5.3 Recoil Separation and Gas Thermalization

1. **Physical Separation**: Use a gas-filled recoil separator (filled with approximately $1\ \mathrm{mbar}$ of helium) to separate the target evaporation residues (EVR)from the massive background beam particles (approximately 10^{12} times the target) via gas-filled magnetic and electric fields. The transmission efficiency must reach $40\% \sim 50\%$.
2. **Gas Thermalization**: The target nuclei enter a gas stopping cell filled with high-purity helium (pressure $50 \sim 100\ \mathrm{mbar}$). The evaporation residues (EVR)decelerate and thermalize into neutral atoms within a few milliseconds. To prevent the evaporation residues (EVR)from adhering to the cell walls, a trace amount of methane must be mixed into the helium, utilizing chemical volatilization to keep the target atoms in the gas phase.

5.4 In-Gas Laser Resonance Ionization

Before the thermalized neutral $Z = 126$atoms are extracted from the gas cell, they must be resonantly excited to Rydberg states and ionized via multi-step laser excitation.

1. **Laser System**: Due to the strong relativistic effects of $Z = 126$, its first excited state transition may lie in the near-infrared or visible band, while the ionization threshold ($6.0\ \mathrm{eV}$) lies in the vacuum ultraviolet (VUV) band. A Ti:Sapphire laser combined with

frequency-doubling crystals, as well as four-wave mixing (FWM), is required to produce tunable laser light.

2. **Scanning Strategy**: Fix one laser and scan the wavelength of another to find resonance peaks (fingerprint identification); then fix the resonance wavelength and scan the laser energy, precisely determining the first ionization energy (IE) by detecting the ion signal threshold.
3. **High Repetition Rate**: The laser repetition rate must be as high as $10\ \text{kHz} \sim 20\ \text{kHz}$ to cover the extremely short residence time of the thermalized atoms.

5.5 Ion Detection and Falsifiability Assessment

- **Background-free detection**: The ionized $Z = 126$ ions are extracted by an electric field and bombard a position-sensitive Microchannel Plate (MCP) detector. Combined with the energy and time signatures of its preceding α-decay chain, single-atom-level zero-background identification is achieved.
- **Criteria**: If the observed first ionization energy $IE \approx 6.0 \pm 1.0\ \text{eV}$ and spectral transitions consistent with the $8p^6 9s^2$ double-closed-shell characteristics appear, it strongly supports the Luoshu geometric model; if the measured value is similar to neighboring elements in the same period (e.g., $IE < 4.5\ \text{eV}$), or if the Rydberg series corresponding to the closed shell cannot be observed, the closed-shell structure hypothesis and the related grid stratification prediction are falsified.

Chapter 6: Light Matrix Topology and Quantitative Verification of the Microscopic Resonance Response Function

This chapter derives the resonance response function of electromagnetic interactions through the topological structure of the Luoshu grid and provides a direct verification protocol at the microscopic scale. This protocol requires no large collider and can be completed in a conventional quantum optics laboratory.

6.1 Topological Metric of the Luoshu Grid and Construction of the Light Matrix

In the Luoshu grid (3×3 topological space), the interaction of energy nodes is strictly described by a 9×9 complex light matrix $\mathbf{L}$. The matrix element L_{ij} represents the transmission amplitude and phase change of energy propagating from node j to node i.

Construction Rules:

1. **Amplitude Attenuation**: Energy propagates along the grid edges; for each step taken (Manhattan distance d_{ij}), the amplitude is multiplied by the golden ratio $\Phi \approx 0.618$.

2. **Phase Rotation**: In the fiber bundle, each step of spiral motion rotates the phase by $\pi/2$. After propagating d_{ij} steps, the phase factor is $e^{i(\pi/2)d_{ij}} = i^{d_{ij}}$.
3. **Raw Transmission Matrix** $\mathbf{T}$: $T_{ij} = \Phi^{d_{ij}} \cdot i^{d_{ij}}$.
4. **Column Normalization**: To satisfy probability/energy conservation, the total energy emitted by each node must be 1; therefore, $\mathbf{L}_{ij} = T_{ij} / \sum_k T_{kj}$.

6.2 Nine Eigenmodes and the Response Function

Strict **numerical diagonalization** of the above $\mathbf{L}$ matrix yields 9 eigenvalues λ_i, which are strictly divided into four categories, perfectly corresponding to the four fundamental interactions:

- **Gravitational Mode (Singlet)**: $\lambda_1 = 1$. Topological origin: the self-loop of the central point 5 to itself ($d = 0$). No intra-layer dissipation ($\Phi^0 = 1$).
- **Strong Force Mode (Four-fold Degenerate)**: $\lambda_2 \approx 0.3266$. Topological origin: the closed loop formed by the four corner points (2,4,6,8).
- **Electromagnetic Force Mode (Four-fold Degenerate)**: $\lambda_3 \approx 0.3849$. Topological origin: the closed loop formed by the four edge midpoints (1,3,7,9).
- **Weak Force Mode (Two-fold Degenerate)**: $\lambda_4 \approx 0.20$. Topological origin: a mixed state of corner points and edge midpoints, leading to parity violation.

Strict Derivation of the Response Function $f(\lambda)$:When the system is continuously driven by an external source, energy accumulates cyclically within the topological mode. This is a geometric series summation process:

$$E_{\text{total}} = E_0(1 + \lambda + \lambda^2 + \ldots) = \frac{E_0}{1 - \lambda}$$

Define the topological response function as:

$$f(\lambda) = \frac{1}{1 - \lambda}$$

Substituting the eigenvalue of the electromagnetic mode $\lambda_3 \approx 0.3849$:

$$f(\lambda_3) = \frac{1}{1 - 0.3849} = \frac{1}{0.6151} \approx 1.626$$

This means: **when the electromagnetic interaction perfectly matches the Luoshu topology, the energy transfer efficiency will exhibit a geometric gain of approximately** 62.6%**.**

6.3 Experimental Verification: Phase Locking at the Atomic and Biological Scales

To verify $f(\lambda_3) = 1.626$, the phase difference must be controlled for blind testing at the microscopic scale.

Experiment 1: Anomalous Gain in Atomic Spectral Oscillator Strength

1. **Theoretical Prediction**: The observed oscillator strength of a specific transition, when the laser phase difference is locked at $\Delta\phi = \pm\pi/2$, should be approximately 1.626times higher than the Standard Quantum Mechanics calculated value.
2. **Operational Protocol**: Use two frequency-locked lasers (one pump, one probe), precisely locking the relative phase difference at $\pm\pi/2$via an Acousto-Optic Modulator (AOM). Use Cavity Ring-Down Spectroscopy (CRDS) to measure the D-line transition absorption peak area of alkali metal atoms (e.g., Rubidium Rb).
3. **Falsification Condition**: If the oscillator strength shows no change during the phase scan, the response function is invalid at the atomic scale.

Experiment 2: Phase Peak of Photosynthesis Quantum Efficiency

1. **Theoretical Prediction**: Chlorophyll molecules are natural nanoscale resonant antennas. When the Carrier-Envelope Phase (CEP) of the excitation light matches the absorption transition phase of Photosystem II (PSII), the quantum efficiency will exceed the traditional thermal dissipation limit, exhibiting an extremum.
2. **Operational Protocol**: Use a femtosecond pulse shaper to control the light field phase, irradiate purified PSII reaction centers, and measure fluorescence decay and quantum yield using Time-Correlated Single Photon Counting (TCSPC) technology.
3. **Falsification Condition**: The quantum efficiency shows no significant fluctuation under any phase.

Chapter 7: Conclusion and Outlook

This document thoroughly specifies the systematic experimental verification protocols for the $8.60\ \mathrm{GeV}$fourth-generation lepton, the superheavy element island of stability, and the microscopic resonance effect.

From the blind scan strategy at Belle II searching for "two high-momentum leptons + large missing transverse momentum," to LHCb utilizing displaced vertices to capture long-lived particles, to laser resonance ionization in a gas stopping cell measuring the $Z = 126$double-closed-shell ionization energy, and to directly testing the 1.626-fold energy amplification effect on a quantum optics platform—we are not merely searching for new particles or new elements, but conducting extremely stringent falsification tests of a completely new underlying geometric principle (the Luoshu Unified Field Theory).

All experimental operational procedures, parameter cutoff thresholds, background estimation methods, and statistical blind scan strategies provided in this document possess extremely high engineering executability. If the above experiments observe the characteristic signals predicted by the theory, it will be a subversive breakthrough against the Standard Model; if the experiments completely exclude these signals, the theory will be strictly boundary-corrected accordingly. Science advances precisely through this cycle of "prediction—verification—correction."

References and Appendix Citations

The underlying particle mass derivations, atomic concentric circle shadow dynamics calculations, and detailed geometric analyses upon which this document is based have been archived as independent appendices on Zenodo. Please consult and cite them directly via the following latest DOIs:

Appendix I: Complete Derivation of the Particle Mass Spectrum and Proton Correction

- **Corresponding Work**: "Verifying Known Particles and Predicting Unknown Particles with Two Constants—Energy Particle Spiral Motion: Experimental Verification Edition"
- **Latest DOI**: **10.5281/zenodo.22782129** (OSF: **10.17605/OSF.IO/7A3MG**)
- *Note: This appendix establishes the mass foundation of the* $8.60\ \mathrm{GeV}$ *fourth-generation lepton and includes the geometric area projection* $(24^{3/2} \times 24^{1/2})$

of the proton mass formula and the two-sided outer-skin fine structure correction $(1 + 2/137)$.

Appendix II: Concentric Circle Shadow Dynamics at the Atomic and Molecular Scales

- **Corresponding Work**: "Concentric Circle Fractals and Shadow Dynamics: A Unified Geometric Framework from Particle Mass to Atomic Ionization Energy" (Volume 29)
- **Latest DOI**: **10.5281/zenodo.22782129** (OSF: **10.17605/OSF.IO/7A3MG**)
- *Note: This appendix details the breathing corrections of prime states and composite states in atomic orbits, as well as the ionization energy predictions for superheavy elements and the* $Z = 126$*island of stability.*

Core Theoretical Foundations (Latest DOIs):

1. **V4 (Geometric Reconstruction)**: Li Shuping. Modern Geometric Reconstruction of the Luoshu Master Equation: Fiber Bundles, Resonance K, and Fractal Center Point 5 (V4.0). DOI: `10.5281/zenodo.23387541`
2. **V3 (Luoshu Master Equation)**: Li Shuping. The Luoshu Master Equation: Breathing, Spirals, and the Unified Dynamics of All Things (V3.0). DOI: `10.5281/zenodo.23232089`
3. **V2 (Spiral Motion Framework)**: Li Shuping. Spiral Motion on the Luoshu Grid: A Geometric Framework from Particle Masses to the Four Fundamental Forces (V2.0). DOI: `10.5281/zenodo.22805463`
4. **Full Series Entry Point**: DOI: `10.5281/zenodo.22782129` or `10.17605/OSF.IO/7A3MG`
